# WILLIAM KIRSCHENMAN

Assistant Professor, Operations Research Department, Naval Postgraduate School (NPS)
5249 Revere Road ◇ Monterey, CA 93940
(910) · 527 · 5377 ◇ william.kirschenman@nps.edu

Updated: 28 October 2026

## EDUCATION

**Ph.D.** Operations Research, *North Carolina State University* *2026*
**M.S.**, Operations Research, *George Mason University* *2020*
**MPA**, Public Administration, *Webster University* *2015*
**B.S.**, Mechanical Engineering, *United States Military Academy* *2010*

## FELLOWSHIPS AND AWARDS

Seth Bonder Scholarship for Applied Operations Research in Military and Security Applications, INFORMS Military and Security Society *2026*
Outstanding Dissertation Award, Department of Industrial and Systems Engineering, NC State University *2026*
Shook Mentor Award, Department of Industrial and Systems Engineering, NC State University *2026*
Phi Kappa Phi Honor Society, North Carolina State University Chapter *2026*
Salah E. Elmaghraby Distinguished Student Award, Operations Research Program, NC State University *2025*
General Omar N. Bradley Research Fellowship in Mathematics *2024, 2025*
Army Operations Research Symposium (AORS) Best Presentation, Wargaming Working Group *2021*
Outstanding Academic Achievement Award in Operations Research, George Mason University *2020*

## PUBLICATIONS

### Refereed Journal Articles

**Kirschenman, W.**, Heese, H.S., Kay, M., King, R., McConnell, B. (2026). Enhancing Military Load Planning: A Prioritized 2-D Orthogonal Packing Approach. *Omega*, Vol. 145, 103638. (Link)

**Kirschenman, W.**, Heese, H.S., Kay, M., King, R., McConnell, B. (2026). The 2-D Orthogonal Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective. *Naval Research Logistics*, 1–22. (Link)

### Trade Journal Publications (Non-Refereed)

**Kirschenman, W.**, Constantino, J., Heese, H.S., King, R., McConnell, B. (2026). Automating Combat Loading to Strengthen LSCO Readiness. *Army Sustainment*, Vol. 58, No. 3, published online July 15, 2026. (Link)

**Kirschenman, W.**, McConnell, B., King, R. (2024). Automated Vessel Selection and Combat Load Planning. *Army Sustainment*, Vol 56, No 4, 58-61. (Link)

## PRESENTATIONS

### Invited Talks

**Kirschenman, W.** (2026). "A Multi-Level Prioritized Packing Framework for Spatial Optimization." Invited seminar, Operations Research Program, Department of Industrial and Systems Engineering, North Carolina State University, Raleigh, NC.

### Conference Presentations

**Kirschenman, W.**, Heese, H.S., Kay, M., King, R., McConnell, B. (2025). "A Prioritized 2-D Packing Model for Military Combat Loading." Presentation at the Annual Meeting of the Institute for Operations Research and the Management Sciences (INFORMS), Atlanta, GA.(Link)

**Kirschenman, W.**, Heese, H.S., Kay, M., King, R., McConnell, B. (2025). "Automating Combat Load Planning using a Prioritized 2-D Packing Approach." Presentation at the 2025 Army Operations Research Symposium (AORS) Symposium, Aberdeen Proving Ground, MD. (Link)

**Kirschenman, W.**, Heese, H.S., Kay, M., King, R., McConnell, B. (2025). "The 2-D Bin Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective." Presentation at the 93rd Military Operations Research Society (MORS) Symposium, National Conference Center, Leesburg, VA. (Link)

**Kirschenman, W.**, Heese, H.S., Kay, M., King, R., McConnell, B. (2025). "Automating Military Load Planning: A Prioritized 2-D Bin Packing Approach." Presentation at the 93rd Military Operations Research Society (MORS) Symposium, National Conference Center, Leesburg, VA. (Link)

**Kirschenman, W.**, Heese, H.S., King, R., McConnell, B. (2025). "The 2-D Bin Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective." Poster presentation, Graduate Student Research Symposium, NC State University, Raleigh, NC. (Link)

**Kirschenman, W.**, Heese, H.S., King, R., McConnell, B. (2024). "Automated Vessel Selection and Combat Load Planning." Presentation at the Annual Meeting of the Institute for Operations Research and the Management Sciences (INFORMS), Seattle, WA. (Link)

**Kirschenman, W.**, Heese, H.S., King, R., McConnell, B. (2024). "Automated Vessel Selection and Combat Load Planning." Presentation at the 2024 Army Operations Research Symposium (AORS) Symposium, Aberdeen Proving Ground, MD. (Link)

**Kirschenman, W.**, Brown, K. (2022). "Distributed Wargaming Visualization and Data Collection." Presentation at the 90th Military Operations Research Society (MORS) Symposium, Marine Corps Base Quantico, VA. (Link)

**Kirschenman, W.**, Brown, K. (2021). "Distributed Wargaming Visualization and Data Collection."Presentation at the 2021 Army Operations Research Symposium (Virtual). (Link)

## MEDIA AND PUBLIC ENGAGEMENT

**Kirschenman, W.** (2026). "Understanding AI: From Chatbots to Agentic Intelligence." Invited guest interview on agentic AI, *Engineering Pack Pod*, College of Engineering, North Carolina State University. Released April 29, 2026. (Link)

## TEACHING AND ACADEMIC SERVICE

**OA3802: Computational Methods for Data Analytics (NPS)** *Fall 2026*
*Instructor; two sections.*

**OA4910: Special Topics: Python Boot Camp (NPS)** *Summer 2026*
*Co-instructor; 5.5-week boot camp.*

**ISE/OR PhD Student Bootcamp (NC State)** *August 7, 2025*
*Invited instructor; one-day Introduction to Python module.* (Materials)

## THESIS ADVISING

*Naval Postgraduate School*

**Capt Samantha Brooker**, U.S. Air Force, Co-advisor *Spring 2027 (in progress)*
*Working Topic: Quantifying Healthcare Delivery Opportunity Cost Of Military Manpower Devoted To Exercises And Deployments*
**MAJ Brandon De La Rosa**, U.S. Army, Second reader *Spring 2027 (in progress)*
*Optimizing The Location Of Unmanned Maritime Logistics Bases In The Pacific*
**ENS Anthony Todd**, U.S. Navy, Second reader *Spring 2027 (in progress)*
**ENS Andrew Vandergrift**, U.S. Navy, Co-advisor *Spring 2027 (in progress)*
*Optimizing Troop And Equipment Load Allocation For Airborne Operations*

## TECHNICAL SKILLS

**Programming Languages:** Julia, Python, R, SQL

**Developer Tools & Version Control:** Git, Subversion (SVN)

**AI Development:** Agentic AI; Machine and Deep Learning (Computer Vision); LLM Abliteration and Multi-GPU API Serving (NVIDIA GB300); PyTorch, CUDA, ROCm; Claude Code, Codex, Cursor.

**Scientific Computing & Optimization:** Mathematical Programming (exact methods, matheuristics, complex solver callbacks, warm-starting, solver parameter tuning), High-Performance Computing (HPC), Distributed & Parallel Computing (CPU/GPU), Bayesian Inference; Proficient with Gurobi, CPLEX, and R Shiny.

**Software Paradigms:** Collaborative Software Development, Object-Oriented Programming (OOP)

## PROFESSIONAL EXPERIENCE

**The Research and Analysis Center (TRAC), Army Futures Command (AFC)** August 2020 - May 2023
*Combat Operations Analyst* *White Sands Missile Range, NM*

- Led the development of TRAC's Visualization Application with Data collection in R (VADR), a cloud-based planning and analytical tool for wargaming and other data collection events.
- Led planning, execution and analysis for AFC's 2020 Capstone Wargame focused on the U.S. European Command area of responsibility.
- Planned and executed warfighter touchpoints, operational unit focus groups, and a strategic commanders assessment for AFC's Project Convergence 22, a Combined Joint Force experiment.

**2nd Infantry Brigade Combat Team, 4th Infantry Division** August 2015 - December 2018
*Company Commander, Brigade Engineer Plans Officer* *Fort Carson, CO*

- **Company Commander:** Commanded two distinct Engineer companies of over 100 Soldiers each, responsible for all training, morale, and equipment valued in excess of $35 million.
- **Brigade Engineer Plans Officer:** Served as the primary engineer planner for a 3,500-Soldier Brigade Combat Team and as a plans officer for Combined Joint Engineer operations in southern Afghanistan.

**G-3 Engineer Section, 2nd Infantry Division** July 2014 - July 2015
*Operations Officer* *Camp Red Cloud, Republic of Korea*

- Planned, synchronized, and assessed engineer operations for the ROK–U.S. Combined 2nd Infantry Division.

**Security Force Advise/Assist Team (SFAAT) #2** December 2012 - September 2013
*Engineer Company Commander Advisor* *Logar Province, Afghanistan*

- Advised an Afghan National Army (ANA) Engineer Company Commander over nine months in all areas of command.

**11th Engineer Battalion, 20th Engineer Brigade** February 2011 - December 2013
*Company Executive Officer, Platoon Leader* *Fort Benning, GA*

- **Company Executive Officer:** Managed operations, training, and maintenance processes for a 198-Soldier Engineer Company.
- **Platoon Leader:** Led a 64-Soldier Engineer Platoon responsible for equipment valued in excess of $20 million.

## PROFESSIONAL MEMBERSHIPS

Military Operations Research Society (MORS) *2021 - Present*
The Institute for Operations Research and the Management Sciences (INFORMS) *2023 - Present*
Engineer Intern in New York *2010*

## SELECTED MILITARY AWARDS

Bronze Star Medal (2), Meritorious Service Medal (2), Army Commendation Medal (2), Army Achievement Medal (5), Military Outstanding Volunteer Service Medal, Steel de Fleury Medal (Army Engineer Association)

## COMMUNITY SERVICE

Cub Scout Leader, White Sands Missile Range, NM *2021 - 2022*
Pianist for Cascade Hills Church (GA) and Joyful Baptist Church (CO) *2011 - 2018*